

Mallika Laasya Ramachandruni

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EDUCATION

University of Wisconsin – Madison

Madison, WI

B.S. Computer Science & Economics

Expected May 2028

- **Relevant Coursework:** Programming 3, Linear Algebra, Statistics, Matrix Methods in Machine Learning

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, HTML/CSS, SQL

Frameworks: React, Next.js, FastAPI, Node.js, Tailwind CSS, scikit-learn

Cloud & Backend: Supabase, Firebase, PostgreSQL, SQLite, REST APIs, Docker

Dev Tools: Git, GitHub Actions, VS Code, CI/CD, Postman

AI/ML: Claude API, Groq API, OpenAI Whisper, scikit-learn, LLM Integration, Prompt Engineering

EXPERIENCE

Undergraduate Learning Center – Incoming CS Tutor

Aug. 2026 – Present

University of Wisconsin – Madison

Madison, WI

- Selected to tutor **CS 200/300** covering core Java fundamentals, OOP inheritance, recursion, ArrayLists, HashMaps, and Big-O time complexity analysis
- Will conduct weekly group office hours and one-on-one sessions, translating complex algorithmic concepts into concrete, debuggable examples for students from varied technical backgrounds
- **Developing targeted supplemental problem sets** addressing common misconceptions in data structures and OOP to support student comprehension beyond lecture material

Pain Context Innovation Lab – Research Assistant

Jan. 2026 – Present

University of Wisconsin – Madison | PI: Dr. Shinye Kim

Madison, WI

- **Reduced average time-to-clinic-match by ~40%** by architecting and deploying the clinic navigation module for a patient-facing web platform serving **100s of chronic pain patients**
- **Cut average user lookup time from 5+ min to under 60 sec** by building responsive, real-time search and filter UI components in **React** with dynamic state management and debounced API calls
- **Integrated the NPPEs NPI REST API** to dynamically query and filter **1M+ clinician records** by insurance type, specialty, and geographic location, surfacing the most relevant providers for each patient
- Collaborated with a **5-person cross-functional team** of clinicians and developers across bi-weekly sprints, managing feature branches and conducting pull request reviews on GitHub

PROJECTS

Vocalis | Next.js, TypeScript, FastAPI, Groq, Whisper, Docker, Firebase [GitHub](#)

June 2026

- **Achieved <2s transcription latency** on responses up to 3 minutes by engineering an end-to-end voice pipeline integrating **OpenAI Whisper** with the Web Audio API for real-time in-browser audio capture and streaming transcription
- **Delivered structured feedback in under 3 seconds** by designing a **scikit-learn scoring model** evaluating interview answers across 4 dimensions: clarity, relevance, depth, and communication quality
- **Built a full-stack AI mock interview platform for 100s of students**, generating role-specific technical and behavioral questions across **3 engineering disciplines** via Groq-hosted LLMs and a RESTful FastAPI backend
- **Containerized the full backend (FastAPI + SQLAlchemy + SQLite) with Docker Compose**; deployed with a Firebase-authenticated CI/CD pipeline enabling reproducible one-command local setup and consistent staging environments

VitalLens | React, TypeScript, Supabase, Claude API, Recharts [GitHub](#)

April 2026

- **Cut patient appointment prep time from 30+ min to under 5 min** by automating personalized doctor prep question generation derived from longitudinal biomarker trends visualized with Recharts across multiple visits
- **Engineered a PDF parsing pipeline** that extracts and classifies **20+ biomarker types** into structured JSON output via **Claude Sonnet 4.6**, serving 100s of users with actionable, plain-English health insights
- **Ensured zero cross-user data exposure** by implementing **Supabase Auth with Row-Level Security** policies and PostgreSQL Edge Functions, achieving HIPAA-aligned per-user data isolation
- **Designed a multi-visit biomarker trend tracking dashboard** enabling patients to monitor health changes over time and arrive at physician appointments with data-driven, targeted questions

CAMPUS INVOLVEMENT

WACM – Undergraduate Activities Chair

Jan. 2026 – Present

University of Wisconsin – Madison

Madison, WI

- **Selected as Undergraduate Activities Chair** for Fall 2026; curate a monthly technical speaker series for **50+ members** featuring engineers from Google, Microsoft, and Amazon

Women in Business Technology – VP of Systems

Jan. 2026 – Present

University of Wisconsin – Madison

Madison, WI

- **Elected VP of Systems** for Fall 2026; architect and maintain club website for a **100+ member organization**, shipping 10+ feature updates per semester and integrating professional networking resources
- **Connected 100+ members with 20+ industry professionals** in business technology through platform-driven outreach and coordinated networking events each semester
- **Honors:** 2nd Place – BadgerAI Hackathon | 3rd Place – WIBT × Uline Case Competition (Oct. 2025) | Java & Python– Certiport (2024)